Assessing the influence of variable annual sampling effort on official national forest inventory statistics

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Abstract

In this study we…

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1. Introduction

The USFS Forest Inventory and Analysis (FIA) Program surveys and analyzes the forests of the United States through a nationwide network of permanent sample locations. These plot locations are distributed across all lands with a spatial sampling intensity of about 1 location per 2400 ha according to an unaligned, quasi-systematic equal probability sample design. This design is spatially dispersed with deviation from a regular grid to avoid alignment of sample locations with periodic population features. Statistical inference on forest population status and change parameters is enabled by measurements of forest vegetation traits made at these locations by field crews.

Historically, FIA sample locations in each state were visited periodically as time, budgets, and workforce capacity allowed. This approach offered logistical flexibility but complicated analysis and reporting timelines. In the early 2000's the program shifted from periodic to annual inventory protocols calling for approximately equal numbers of plot measurements each year on schedules intended to be fixed and consistent within states. Under this annual inventory, each plot location in a state is assigned to one of a set of panels, and all plot locations assigned to a panel are designated for measurement in the same year.

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The panels in a state together comprise an inventory cycle, and inventory cycles range in length from 5 or 7 years in the eastern US to 10 years in the west. Once the final panel in an inventory cycle is complete, remeasurements at sample locations in the first panel begin the following year. For more information regarding the FIA sample design see [1, 2, Chapter 2].

In general, FIA requires that a panel be complete before the associated tabular data are pushed to its public-facing databases, which the program endeavors to do within 1 year of panel completion. Similarly, FIA requires a complete inventory cycle within a state to publish official state-scale estimates of forest parameters in fulfillment of Congressional reporting obligations. In practice, however, adherence to the annual design described above has proven very challenging or impossible for the FIA Program. Instead, delayed and skipped field measurements cause unintentional lengthening of inventory cycles and/or survey nonresponse [see, e.g., 3], at rates that vary regionally and over time.

The underlying causes of delayed and skipped field measurements are varied, and can be chronic (e.g., denied access to private lands [4], logistical or administrative challenges in remote regions [5], persistent workforce and budgetary constraints [6]) or can result from discrete events (e.g., the COVID19 pandemic [7]). Late measurements of temporally dynamic variables can induce a "time lag" bias in point estimates when the time component of the estimand is not adjusted accordkingly. Further, survey nonresponse can introduce nonresponse bias to estimates of population parameters when response propensity is correlated with the attribute of interest. Better understanding of patterns in late and skipped measurements [6], and the investigation of new sampling and estimation techniques to ameliorate the statistical consequences of nonresponse [8, 9] are active research topics, but beyond the scope of the present study.

Of immediate practical concern for FIA and its external partners are the strain on field staff and delays to tabular data publication that result from delayed field measurements. Methods have been proposed to control the measurement delay process by disaggregating FIA's annual panel structure into subpanels while still preserving the spatial dispersion of the FIA sample network within field season years. Disaggregation of panels into subpanels enables intentional alignment of workloads with anticipated data acquisition workforce capacity in each field season year through the application of variable annual field sampling effort within individual states.

Though the logistical advantages of variable annual effort (VAE) are profound (i.e., it can help ensure the long-term viability of the inventory in the context of funding and staffing volatility), the statistical consequences of the redistribution of partial panels within inventory cycles, and of extended and variable inventory cycle lengths, are unknown across states and epochs. In particular, a need exists to understand and communicate to stakeholders how adoption of VAE protocols may influence the statistical properties of forest parameter estimates published by FIA in fulfillment of Congressional reporting imperatives.

Therefore, our research objectives in this study are to:

1. define three VAE regimens with respect to subpanels (fast and slow start, dropped subpanels, and extended and shortened cycle lengths) intended to be representative of circumstances FIA could conceivably encounter in program administration
2. simulate the influence of those three VAE regimens on state-scale, inventory-cycle wide estimates of total net merchantable timber volume for three US states with distinct forest types, ecological attributes, and FIA inventory cycle lengths.
3. visualize and discuss simulation results in the context of implications for FIA program operations and the management of US forests.

2. Data & Methods

2.1. Study Area

3 states…

2.2. FIA Data

Data from those 3 states…

2.3. Initial Notation

2.3.1. Sample locations

Let

$$\mathbf{s}_i = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

be the bivariate spatial coordinate vector for the ith FIA sample location in a focal US state, $i = 1, \dots, n$, where n is the number of all sample locations in the focal state, and x_1 and x_2 are measures of longitude and latitude respectively.

2.3.2. Micropanels

Let

$$\mathcal{Q} = \{q_r : \ r = 1, \dots, R\}$$

be a set of "micropanels" q_r that partition the n sample locations into approximately equivalently-sized subsets, such that each $q_r \in \mathcal{Q}$ achieves approximate spatial dispersion across the focal state of each of the s_i it contains. The q_r defined here are meant to correspond to collections of sample locations referred to in the FIA Program as "Panel 70's", which imply $R = 70$, or "Panel 140's", which imply $R = 140$.

In CONUS states where the FIA Program is administered by the Pacific Northwest Research Station, $R = 140$ in present day inventory operations. In

the rest of the CONUS states, $R = 70$. Though this notation is intended to be flexible enough to accommodate a range of possible values of R, in this study we don't consider PNWRS states, and so we'll only consider $R = 70$ for practical purposes.

2.3.3. Macropanels

The R micropanels can be further aggregated into the set of M macropanels

$$\mathcal{P} = \{P_m : \ m = 1, ..., M\}$$

for any value of M such that $M|R$. That is, R is evenly divisible by M, and so the macropanels $P_m \in \mathcal{P}$ are perfectly evenly partitioned by the micropanels $q_r \in \mathcal{Q}$.

In the context of FIA, each $P_m \in \mathcal{P}$ can be thought of as an "annual panel", meaning that each $s_i \subset q_r \subset P_m$ is scheduled to be visited by a field crew in year in the tth year in a chronological series of years indexed in the set

$$\mathcal{T} = \{t : t = 1, ..., T\}$$

where $T = M$. Therefore, $\mathcal{P}$ can be thought of as in inventory cycle, of cycle length M, composed of the annual panels $P_m \in \mathcal{P}$, which themselves are evenly subdivided into the micropanels $q_r \in \mathcal{Q}$.

$M = T$ here is intended to correspond to the common state-level inventory cycle lengths prescribed in the FIA program, and so in this study $M \in \{5, 7, 10, 14\}$. Because the micropanels are initially evenly distributed across the annual panels, the number of micropanels per macropanel can be denoted by k as

$$k = \frac{R}{M}$$

A general expression to recover the identities of the micropanels contained in any macropanel P_m is given by

$$P_m = \{q_r \in \mathcal{Q} : (m-1)k \ + \ 1 \leq r \leq mk\}$$

2.3.4. Macropanel Lookup Function

Conversely, to determine the identity of the macropanel to which micropanel r belongs, we can define a macropanel lookup function

$$f(r) = \left\lceil \frac{r}{k} \right\rceil = \left\lceil \frac{r \cdot M}{R} \right\rceil$$

where the ceiling function $\lceil x \rceil$ returns the smallest integer not less than x, and $f(r) = m$ outputs the macropanel index ($1 \leq m \leq M$).

2.4. VAE Scenarios

We will now extend this notation to describe in general terms three VAE scenarios achieved by either i) reassigning micropanels to different macropanels, or ii) dropping fixed numbers of micropanels per macropanel over the course of an inventory cycle.

2.4.1. Slow/Fast Start

2.4.1.1. Modified Partition Notation (Shifted Configurations).

For an uneven or shifted layout where subpanels are structurally rearranged, the uniform mapping breaks down. To formalize this, we introduce an explicit boundary sequence vector $\mathbf{B}$ containing $M + 1$ elements:

$$\mathbf{B} = (b_0, b_1, \dots, b_M)$$

For the specific modified configuration where $M = 5$, the boundaries are explicitly assigned as:

$$\mathbf{B} = (0, 10, 20, 38, 56, 70)$$

Each individual macro panel P_i (for $i = 1, 2, \dots, M$) is then constructed by pulling indices between adjacent elements of the boundary sequence:

$$P_i = \{q_j \in \mathcal{Q} \mid b_{i-1} + 1 \leq j \leq b_i\}$$

2.4.1.2. Explicit Macro Panel Contents.

- $P_1 = \{q_j \in \mathcal{Q} \mid 1 \leq j \leq 10\}$
- $P_2 = \{q_j \in \mathcal{Q} \mid 11 \leq j \leq 20\}$
- $P_3 = \{q_j \in \mathcal{Q} \mid 21 \leq j \leq 38\}$
- $P_4 = \{q_j \in \mathcal{Q} \mid 39 \leq j \leq 56\}$
- $P_5 = \{q_j \in \mathcal{Q} \mid 57 \leq j \leq 70\}$

2.4.1.3. Modified Panel Lookup Function Alternatives.

Because the subpanel distributions are no longer uniform, the scalar ceiling function cannot be used. Below are two equivalent mathematical approaches to express the lookup function $f(j)$ for the shifted configuration.

2.4.1.4. Approach 1: The Summation / Indicator Function Approach.

We can define the lookup function $f(j)$ by counting how many boundaries the subpanel index j has successfully crossed. Using the Iverson bracket notation (where the brackets evaluate to 1 if the enclosed condition is true, and 0 if false), the macro panel index can be calculated directly by checking membership across all intervals:

$$f(j) = \sum_{i=1}^{M} i \cdot [b_{i-1} + 1 \leq j \leq b_i]$$

Alternatively, an even more concise formulation counts how many upper bounds b_i from our sequence $\mathbf{B} = (0, 10, 20, 38, 56, 70)$ are strictly less than j, then offsets the result by 1:

$$f(j) = 1 + \sum_{i=1}^{M-1} [j > b_i]$$

2.4.1.5. Example Evaluation.

To locate subpanel q_{25} using the upper bounds $b_1 = 10, b_2 = 20, b_3 = 38, b_4 = 56$:

$$f(25) = 1 + [25 > 10] + [25 > 20] + [25 > 38] + [25 > 56]$$
$$f(25) = 1 + 1 + 1 + 0 + 0 = 3$$

This correctly maps subpanel q_{25} to Macro Panel P_3.

3. Discussion

4. Conclusion

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